

**CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY**

Fourth Semester of B.Tech. Theory (IT) Examination

November 2013

MA206 Probability, Statistics &amp; Numerical Analysis

Date: 15/11/2013, Friday Time: 01:30 p.m. to 04:30 p.m. Maximum Marks: 70

**Instructions:**

1. This question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever if required.
4. Use of scientific non-programable calculator is allowed.

**SECTION I**

Q.1. Wright whether following statements are true or false. [07]

- (a) The arithmetic mean of four values of variable  $x$  is 8, if  $\sum_{i=1}^3 x_i = 28$ , then  $x_4 = 4$ .
- (b) The number of different orders in which five letters  $a, b, c, d, e, f$  can be arranged so as to begin with  $ab$  is  $\binom{6}{2}$ .
- (c) If  $A$  and  $B$  are two events on sample space  $S$  with probability function  $P(\cdot)$ , then  $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ .
- (d) If a random variable  $X$  has Binomial probability distribution with  $n = 6$  and  $p = 0.20$ , the mean of distribution is 0.96.
- (e) If  $x_1, x_2, \dots, x_n$  denotes random sample of size  $n$ , then sample Variance is given by  $s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$ .
- (f) The value of Karl Pearson coefficient of correlation  $r$  satisfies  $0 \leq r \leq 1$ .
- (g) The value of rank correlation coefficient between  $n$  pairs of ranks

$$(x_i, y_i), i = 1, 2, \dots, n \text{ is given by } r_R = 1 - \frac{6 \sum_{i=1}^n (x_i - y_i)^2}{n(n^2 - 1)}.$$

Q.2. Answer the following questions. [14]

- (a) The following frequency distribution show the monthly fees (in dollars) paid by a random sample of 49 users of commercial Internet Service Providers in August 2000. [03]

|                     |          |           |           |           |           |
|---------------------|----------|-----------|-----------|-----------|-----------|
| Monthly fees:       | [5 - 13) | [13 - 21) | [21 - 29) | [29 - 37) | [37 - 45) |
| Number of Students: | 5        | 23        | 16        | 3         | 2         |

Represent the frequency distribution by Histogram.

- (b) Write important properties of arithmetic mean. [04]
- (c) The following are the number of automobile accidents that occurred at 60 major intersections in a certain city during the Fourth of July weekend: [07]

0 2 5 0 1 4 1 0 2 1 5 0 1 3 0 0 2 1 3 1  
 1 4 0 2 4 1 2 4 0 4 3 5 0 1 3 6 4 2 0 2  
 0 2 3 0 4 2 5 1 1 2 2 1 6 5 0 3 3 0 0 4

Group these data into frequency distribution showing how often each of the values occurs. Also compute the mean and median number of automobile accident using frequency distribution.

OR

Q.2. Answer the following questions.

[14]

- (a) The following data pertain to the number of computer jobs per day and the central processing unit (CPU) time required. [07]

|                    |   |   |   |   |    |
|--------------------|---|---|---|---|----|
| Number of jobs $x$ | 1 | 2 | 3 | 4 | 5  |
| CPU time $y$       | 2 | 5 | 4 | 9 | 10 |

- Construct the scatter plot of the data.
  - Use the equation of the least squares line to estimate the mean CPU time at  $x = 3.5$
- (b) A company markets and repairs small computers. To study the relationship between the length of a service call and the number of electronic components in the computer that must be repaired or replaced, a sample of records on service calls was taken. The data consist of the length of service calls in minutes and the number of components repaired. The data are presented in Table below. [07]

|             |    |    |    |    |    |    |    |
|-------------|----|----|----|----|----|----|----|
| Minutes $y$ | 23 | 29 | 49 | 64 | 74 | 87 | 96 |
| Units $x$   | 1  | 2  | 3  | 4  | 4  | 5  | 6  |

- Construct the scatter plot of the data.
- Compute Karl Pearson coefficient of correlation  $r$  using data in table.

Q.3. Answer the following questions.

[14]

- (a) Assume that the number of messages input to a communication channel in an interval of duration  $t$  seconds is Poisson distributed with parameter  $0.3t$ . Compute the probabilities of the following events: [04]

- Exactly three messages will arrive during a 10 second interval.
  - At most 2 messages arrive in a period of 20 second.
- (b) State the assumptions of Binomial distribution. It is known that the probability that an item produced by a certain machine will be defective is 0.1. If a sample of eight items are selected at random from the output of the machine, determine the probability that a sample will contain, [05]
- No more than one defective item.
  - Exactly two defective items.

- (c) In an intelligence test administered to 1000 students the average score was 42 and standard deviation 24. Assuming that the distribution of score is approximately normal Find probability of score of student [05]

- exceeding 50
- lying between 30 and 54.

You are given  $P\{Z \leq 0.33\} = 0.6293$ ,  $P\{Z \leq 0.5\} = 0.6915$

OR



Q.3. Answer the following questions. [14]

- (a) Define Uniform distribution. Determine the mean and variance of Uniform distribution. If a random variable  $X$  is uniformly distributed over  $(0,10)$ , calculate the probability  $P\{3 \leq X \leq 8\}$  [07]
- (b) Define Exponential distribution. Determine the mean and variance of exponential distribution. The time (in hours) required to repair a machine is an exponentially distributed random variable with parameter  $\mu = 0.5$ . What is the probability that a repair time exceeds 2 hours? [07]

## SECTION II

Q.4. Write whether following statements are true or false. [07]

- (a) The Newton Raphson formula to find real root of algebraic equation is

$$x_{i+1} = x_i - \frac{f'(x_i)}{f(x_i)}, i = 1, 2, \dots$$

- (b) If  $f(x)$  is real and continuous in the interval  $(a, b)$  and  $f(a)$  and  $f(b)$  have opposite signs, that is  $f(a)f(b) < 0$  then there is at least one real root between  $a$  and  $b$ .
- (c) With usual notations one can write  $\Delta^2 y_0 = y_2 - 2y_1 + y_0$
- (d) Assume we have a table of values  $(x_i, y_i)$ ,  $i = 0, 1, 2, 3, 4$  of a function  $y = f(x)$  and the values of  $x$  are equally spaced. The entry in Forward difference table correspond to  $\Delta^5 y$  is zero.
- (e) Assume we have a table of values  $(x_i, y_i)$ ,  $i = 0, 1, 2, 3$  of a function  $y = f(x)$  and the values of  $x$  are equally spaced. By using these table values, we can determine Newton's forward difference interpolation polynomial with degree 3.
- (f) The relationship between shift operator  $E$ ,  $\Delta$  is  $E \equiv \Delta + 1$ .
- (g) Assume we have a table of values  $(x_i, y_i)$ ,  $i = 0, 1, 2$  of a function  $y = f(x)$ . The degree of Lagrange Interpolating polynomial determined from these values is 3.

Q.5. Answer the following questions. [14]

- (a) Given the table of values [07]

|                |        |        |        |        |
|----------------|--------|--------|--------|--------|
| $x$            | 150    | 152    | 154    | 156    |
| $y = \sqrt{x}$ | 12.247 | 12.329 | 12.410 | 12.490 |

evaluate  $\sqrt{155}$  using Lagrange's interpolation formula.

- (b) Find a root of the equation  $x \sin x + \cos x = 0$  using Newton Raphson method. [05]  
Perform four iterations starting with  $x_0 = 3.1416$
- (c) Define Absolute error, and relative error in numerical analysis. [02]

OR

Q.5. Answer the following questions.

[14]

- (a) Determine the constants  $a$  and  $b$  by least-squares method such that  $y = ae^{bx}$ , [07]  
fits the following data.

|     |        |        |         |        |
|-----|--------|--------|---------|--------|
| $x$ | 1.0    | 1.2    | 1.4     | 1.6    |
| $y$ | 40.170 | 73.196 | 133.372 | 243.02 |

- (b) Find a real root of the equation  $x^3 - 4x - 9 = 0$  using false position method, [05]  
in the interval  $[2,3]$ . Perform four iterations.
- (c) What is floating point representation of real number in digital computer? [02]

Q.6. Answer the following questions.

[14]

- (a) From Taylor series for  $y(x)$ , find  $y(0.1)$  correct to four decimal places if  $y(x)$  [04]  
satisfies

$$y' = \frac{dy}{dx} = x - y^2 \text{ and } y(0) = 1$$

- (b) Given  $\frac{dy}{dx} = y - x$  where  $y(0) = 2$ , Find  $y(0.1)$  and  $y(0.2)$  correct to four [05]  
decimals using the *fourth order Runge-Kutta method*.

- (c) Use Simpson 1/3 rule with 11 points to compute an approximation to the [05]  
integral  $\int_1^6 f(x)dx$ . Where  $f(x) = 2 + \sin(2\sqrt{x})$ .

OR

Q.6. Answer the following questions.

[14]

- (a) Use two point Gaussian integration formula to approximate  $\int_{-1}^1 \frac{1}{x+2} dx$ . [05]

Use following table values.

|     |              |       |
|-----|--------------|-------|
| $n$ | $\pm u_i$    | $W_i$ |
| 2   | 0.5773502692 | 1.0   |

Also determine relative error of integration.

- (b) Solve the equation  $\frac{dy}{dx} = x + y^2$ , where  $y(0) = 1$ , using Picard's method [05]

- (c) Determine the value of  $y$  when  $x = 0.1$  given that [04]

$$y(0) = 1 \text{ and } y' = \frac{dy}{dx} = x^2 + y$$

by using Euler's formula. Take  $h = 0.05$ .